

A PROJECT TITLE
ON
***Portfolio Risk Analysis and Investment Recommendation System Using
PCA***

Project submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
in
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by

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DEPARTMENT OF CS&IT

Portfolio Risk Analysis and Investment Recommendation System Using PCA

1. Abstract:

The increasing complexity of financial markets has made portfolio risk assessment and investment decision-making more challenging. Investors often rely on historical market data to evaluate the performance and risk associated with different financial assets. This project proposes a **Portfolio Risk Analysis and Investment Recommendation System Using Principal Component Analysis (PCA)** that applies machine learning techniques to analyse stock market data, identify major risk factors, and generate investment recommendations.

The system collects historical stock market data from publicly available financial sources and performs data preprocessing, including missing value handling, feature scaling, and feature engineering. Principal Component Analysis (PCA) is then applied to reduce the dimensionality of the dataset while preserving the maximum amount of information. This simplifies the analysis by identifying the most significant factors affecting portfolio performance.

In addition to PCA, the project incorporates supervised and unsupervised machine learning techniques to evaluate portfolio risk and classify investment opportunities. Clustering methods are used to group similar stocks based on their characteristics, while classification models assist in predicting portfolio risk levels. The performance of different machine learning models is evaluated using appropriate metrics such as accuracy, precision, recall, F1-score, and cross-validation.

The proposed system provides investors with graphical visualizations of portfolio performance, risk distribution, and principal components, along with investment recommendations aimed at improving portfolio diversification and reducing overall risk. The project also demonstrates the complete machine learning lifecycle, including data collection, preprocessing, model training, evaluation, and prediction.

This project offers a practical application of machine learning concepts in the financial domain and provides an effective decision-support tool for portfolio analysis while demonstrating modern data-driven investment strategies.

2. INTRODUCTION:

Financial markets generate large volumes of data every day, making investment analysis increasingly complex. Investors require efficient tools to identify risk, understand market behaviour, and make informed investment decisions. Traditional portfolio analysis often relies on statistical methods that become difficult to interpret when handling multiple financial variables.

Machine learning provides efficient techniques for analysing large financial datasets and identifying hidden patterns. Principal Component Analysis (PCA) is one such dimensionality reduction technique that extracts the most significant information from complex datasets while reducing computational complexity.

The proposed system combines PCA with machine learning algorithms to analyse portfolio risk, group similar stocks, evaluate investment performance, and provide meaningful investment recommendations. By integrating data preprocessing, feature engineering, model evaluation, and visualization, the project demonstrates a complete machine learning workflow suitable for financial data analysis.

3. SOFTWARE REQUIREMENTS:

Operating System: Windows 10/11
Programming Language: Python 3.13+
IDE: Visual Studio Code

Libraries:

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- yfinance
- Joblib

Version Control: Git & GitHub